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Marks 27.00/27.00

Grade 10.00 out of 10.00 (100%)

Question 1

Complete

Mark 11.00 out of 11.00

Induction Motor Calculations

1. Determine the synchronous speed of the rotating magnetic field of a three-phase, 10-pole induction motor when the supply frequency is:

a) 50 Hz

Answer: rpm

(Enter your answer as a whole number)

b) 25 Hz

Answer: rpm

(Enter your answer as a whole number)

2. Determine the number of poles of a three-phase induction motor when the stator rotating magnetic field rotates at 250 rpm and the supply frequency is 50 Hz.

Answer:

(Enter your answer as a whole number)

3. The supply frequency is 50 Hz. The shaft speed of a three-phase induction motor is 1450 rpm.

Determine:

- the number of poles
- the relative slip

Number of poles:

(Enter your answer as a whole number)

Relative slip:

%

(Enter your answer with two decimal places)

4. The supply frequency is 50 Hz and the motor has 12 poles.

Determine the synchronous speed of the rotating magnetic field.

Answer:

rpm

(Enter your answer as a whole number)

5. A four-pole **induction motor** supplied with 50 Hz has a relative slip of 4%.

Determine the shaft speed.

Answer:

rpm

(Enter your answer as a whole number)

6. The nameplate data of a three-phase **induction motor** are:

Rated power $P_N = 1.5 \text{ kW}$

Rated voltage $U_N = 380 \text{ V}$

Rated current $I_N = 3.7 \text{ A}$

Power factor $\cos\varphi = 0.79$

Frequency $f = 50 \text{ Hz}$

Speed $n = 1420 \text{ rpm}$

Calculate:

a) Efficiency η

Answer:

(Enter your answer with three decimal places)

b) Active current (in-phase current)

Answer:

A

(Enter your answer with two decimal places)

c) Reactive current component

Answer:

A

(Enter your answer with two decimal places)

7. A three-phase induction motor is connected to a power supply where the phase voltage is 220 V and the frequency is 50 Hz.

The motor current is 5 A and the power factor is 0.85.

Calculate the active power drawn from the supply.

Answer:

kW

(Enter your answer with one decimal place)

Question 2

Complete

Mark 16.00 out of 16.00

Induction Motor Calculations and Understanding

1. Operating principle

A balanced three-phase current in the stator winding produces a .

The interaction between rotor current and stator magnetic field produces .

2. Rotating magnetic field

The phase displacement between the three currents of a balanced three-phase system is electrical degrees.

In a balanced three-phase system the magnitude of the resultant magnetic field vector remains approximately .

3. Phase sequence

If any two supply phases are exchanged, the direction of rotation of the magnetic field .

4. Synchronous speed

Calculate the synchronous speed of a four-pole [induction motor](#) supplied with 50 Hz.

Answer: rpm

(Enter your answer as a whole number)

5. Slip

A four-pole, 50 Hz [induction motor](#) operates at a shaft speed of 1440 rpm. Calculate the relative slip.

Answer: %

(Enter your answer with two decimal places if necessary)

6. Why is slip necessary?

When shaft speed equals synchronous speed, rotor voltage and rotor current become .

Therefore electromagnetic torque becomes .

7. Star and Delta connection

In a star connection:

$$U_{\text{phase}} = U_{\text{line}} / \sqrt{3}$$

In a delta connection:

$$U_{\text{phase}} = U_{\text{line}}$$

8. Power, efficiency and losses

A motor receives 10.0 kW electrical input power and delivers 9.0 kW shaft output power. Calculate the efficiency.

Answer: %

(Enter your answer with one decimal place)

The difference between electrical input power and shaft output power becomes motor .

9. Torque and voltage

Available **induction motor** torque is approximately proportional to the square of the voltage. If the voltage falls to 90% of nominal voltage, the available torque becomes approximately

Answer: %

of the original torque.

10. Starting and speed control

Which method directly controls both frequency and voltage and therefore enables continuous speed control?

For an ideal centrifugal fan or pump:

$$P_2 / P_1 \approx (n_2 / n_1)^3$$

If speed is reduced to 80 %, power demand becomes approximately

Answer: %

of the original value.

Pass requirement: Minimum score required: **8/10**.